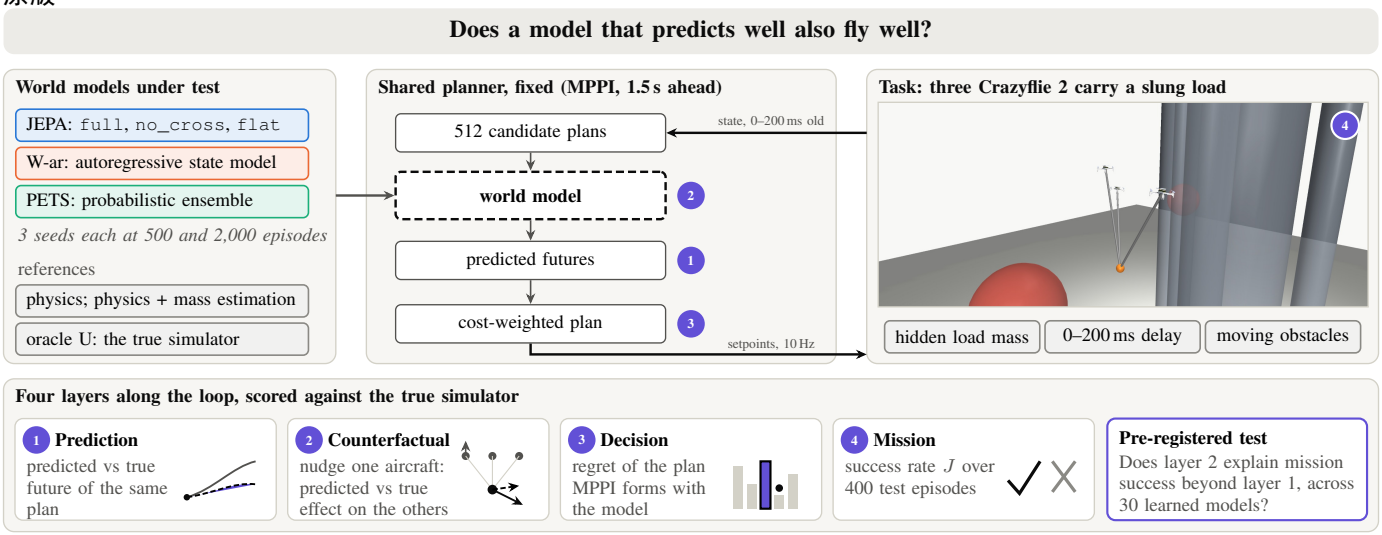
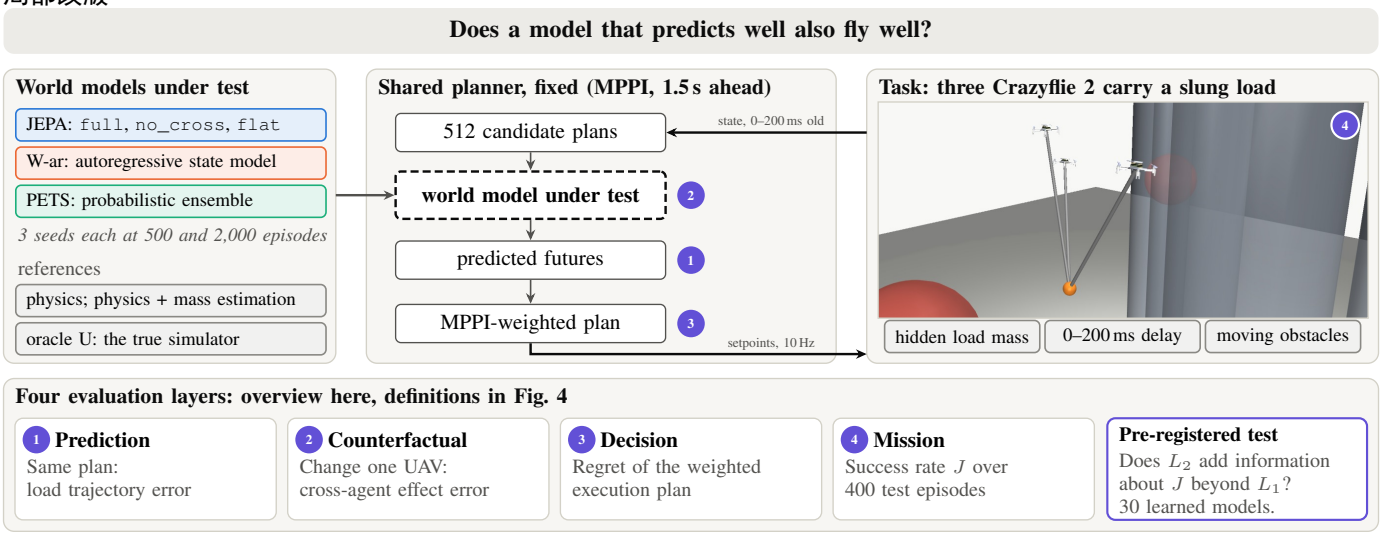
Fig. 1 局部修改审核

基于原版代码，保留原配色和字体家族，仅适配字号及布局。512 candidates、训练数据量和种子数量本轮保留。Overleaf 未修改。

原版



局部改版

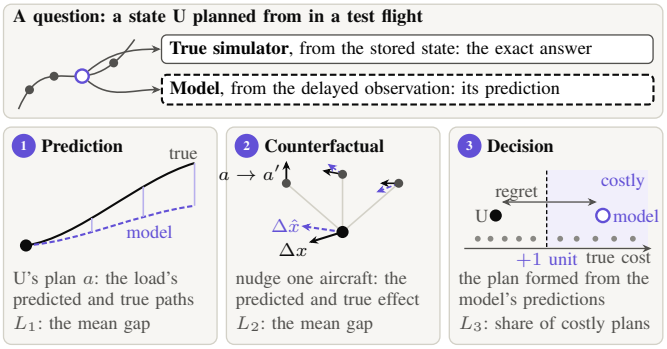


底部仅保留四层的简要概述，细节交给 Fig. 4；场景使用原图裁切，没有生成或修补图像。

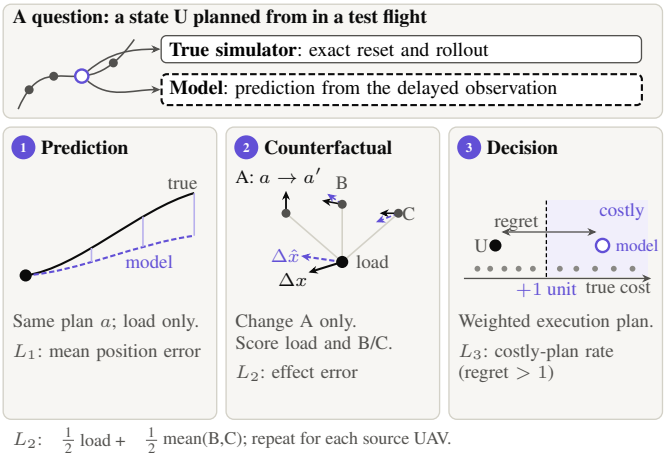
Fig. 4 关联调整审核

保留原版横向三面板和黑实线/紫虚线编码。增加图形与文字间距，明确 L2 的评分对象和 L3 的加权执行计划。

原版



关联改版

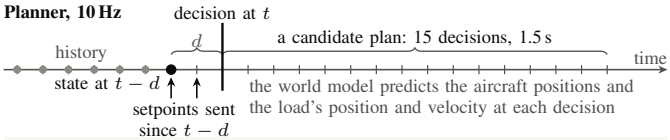


L2：只扰动 A，评分载荷及 B/C，再轮换扰动来源。L3：评估 MPPI 加权形成的执行计划，不是候选中的最优点。所有轨迹和成本图形均为原理示意。

Fig. 3 局部修改审核

基于原版源码。保留原配色和字体家族；仅调整字号、布局与信息层次。Overleaf 未修改。

原版



Onboard, 500 Hz: Lee controllers on the true state; setpoints ramp over 0.1 s

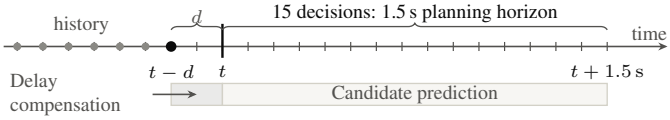
Observed: goal, pillars, setpoints sent; from $t-d$, the aircraft, load and sphere states.
Never observed: load mass, cable tensions, controller and motor states.

局部改版

Available information

Delayed: aircraft, load and moving-obstacle states.
Known: goal, pillars, sent setpoints and observation history.

World-model prediction



Use sent setpoints to bridge $t-d \to t$.

Predict aircraft positions and load position/velocity.

Execution

Planner: 10 Hz; onboard Lee control: 500 Hz on the true state.
Setpoints ramp to their new values over 0.1 s.

Hidden: load mass, cable tensions, controller and motor states.

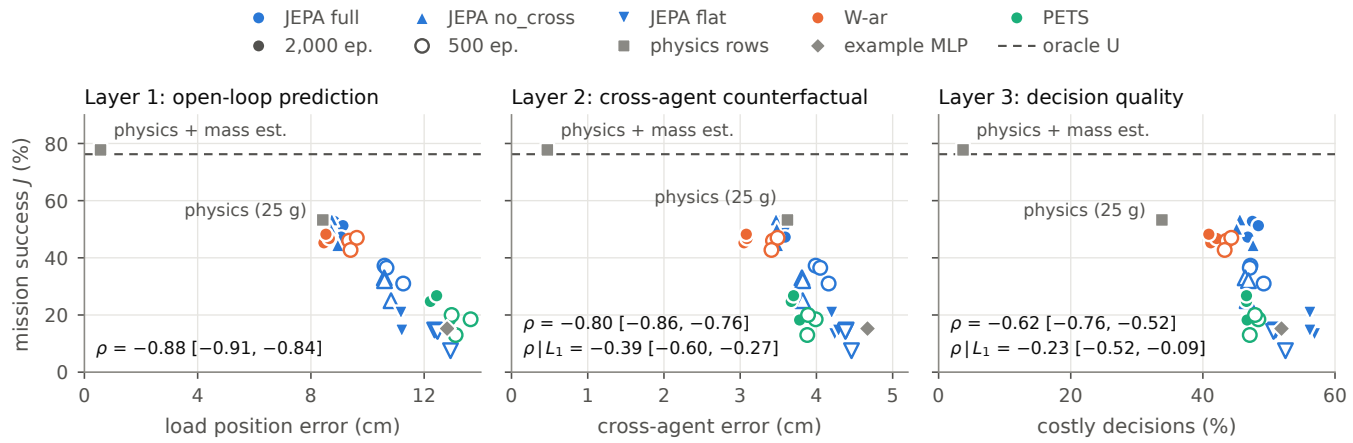
Oracle U uses true current state; time to scale, delay shown at 200 ms.

分为可用信息、世界模型预测、控制执行三个区域。延迟补偿与未来预测分开，保留原时间比例、15 步/1.5 秒、10 Hz/500 Hz、0.1 秒 ramp 及 oracle 例外。

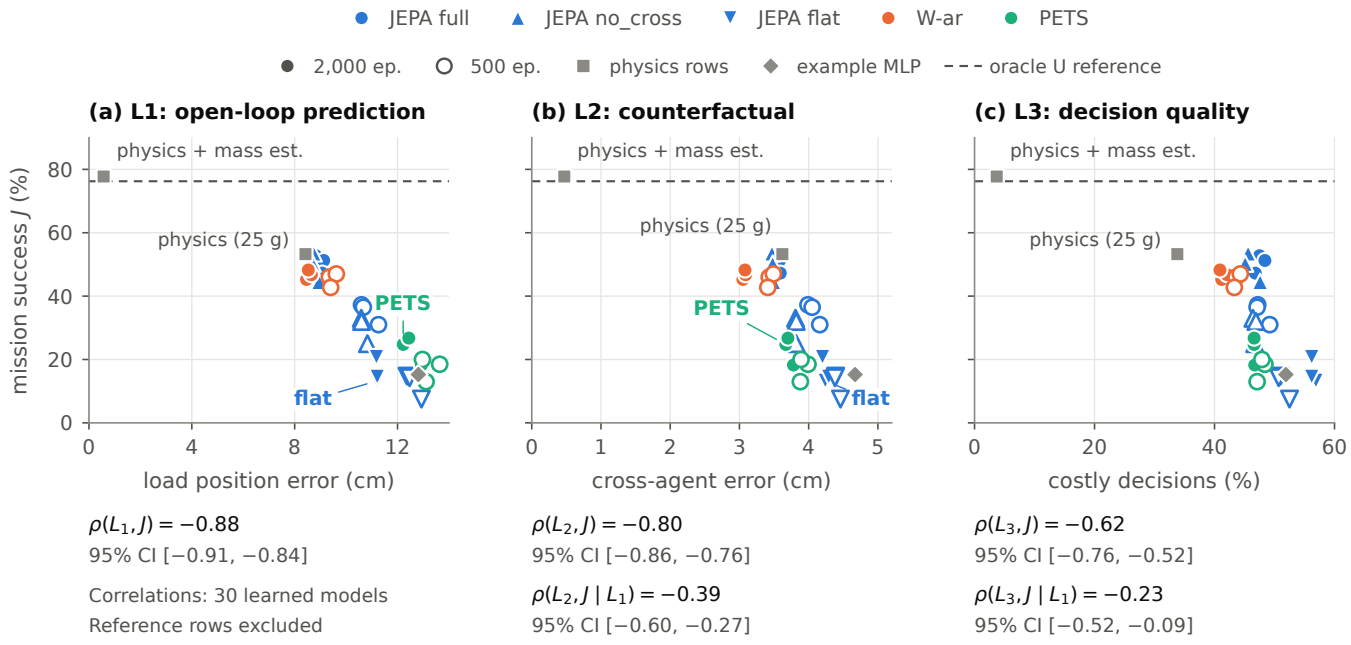
Fig. 5 局部修改审核

基于原版绘图代码，保留原配色、DejaVu Sans 字体家族、所有点位及统计数值。Overleaf 未修改。

原版



局部改版



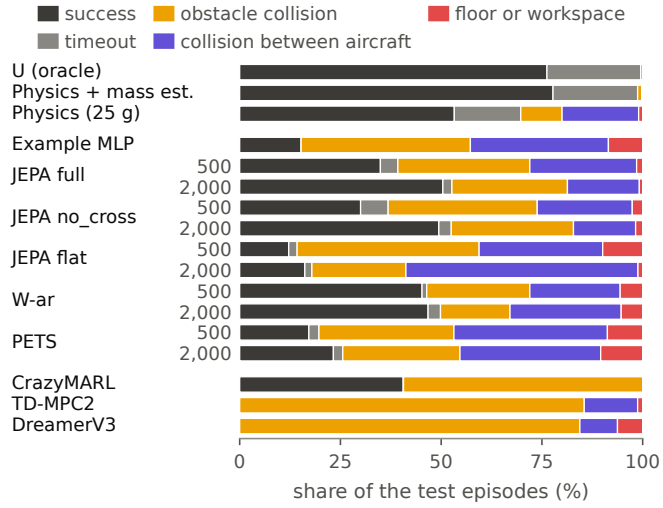
增加面板编号；模型与数据量图例分行；增大标记及文字；对 PETS/flat 增加标签。统计量移到各面板下方，普通相关与控制 L1 后的偏相关分开。

每个面板均保留 30 个待分析模型、两个 physics 参考点及一个示例 MLP；oracle 虚线是参考表现，不是严格上限。全部点位保持不变。

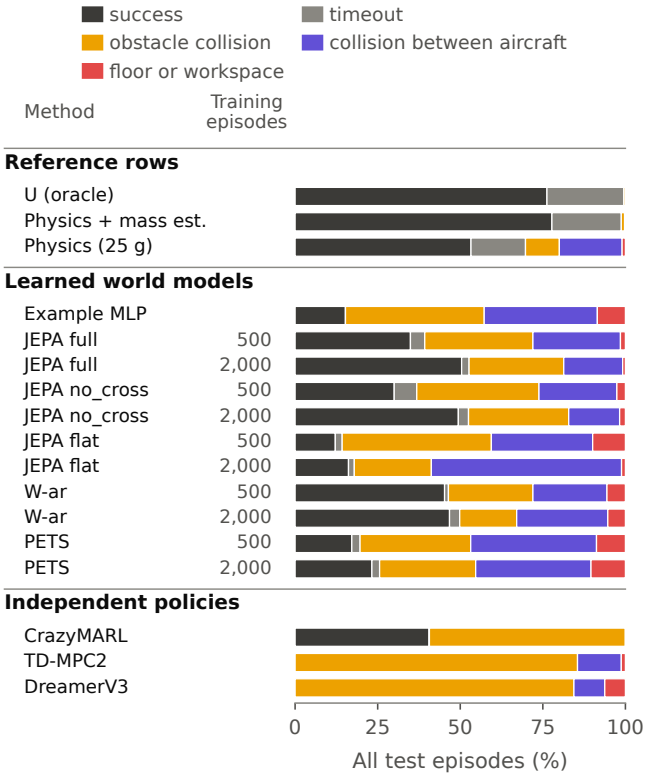
Fig. 6 局部修改审核

基于原版 plot_outcomes.py，保留原配色、DejaVu Sans 字体及全部计数。Overleaf 未修改。

原版



局部改版

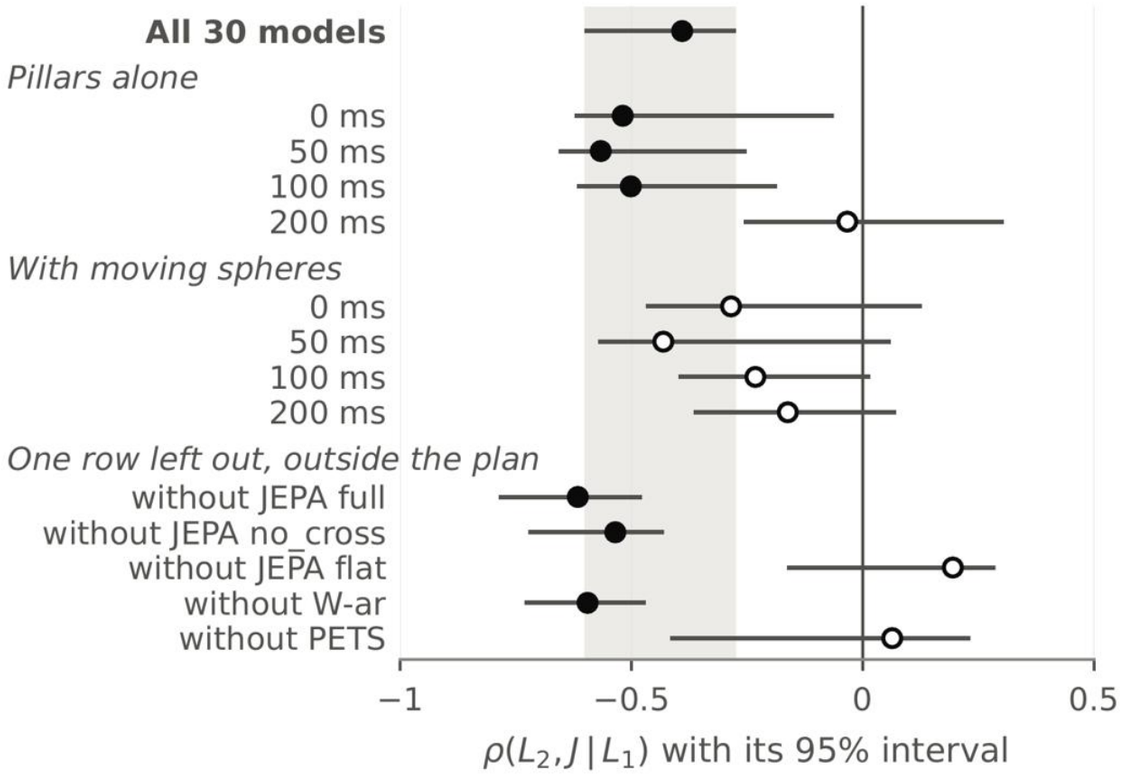


400 flights per model; 1,200 for three pooled seeds.

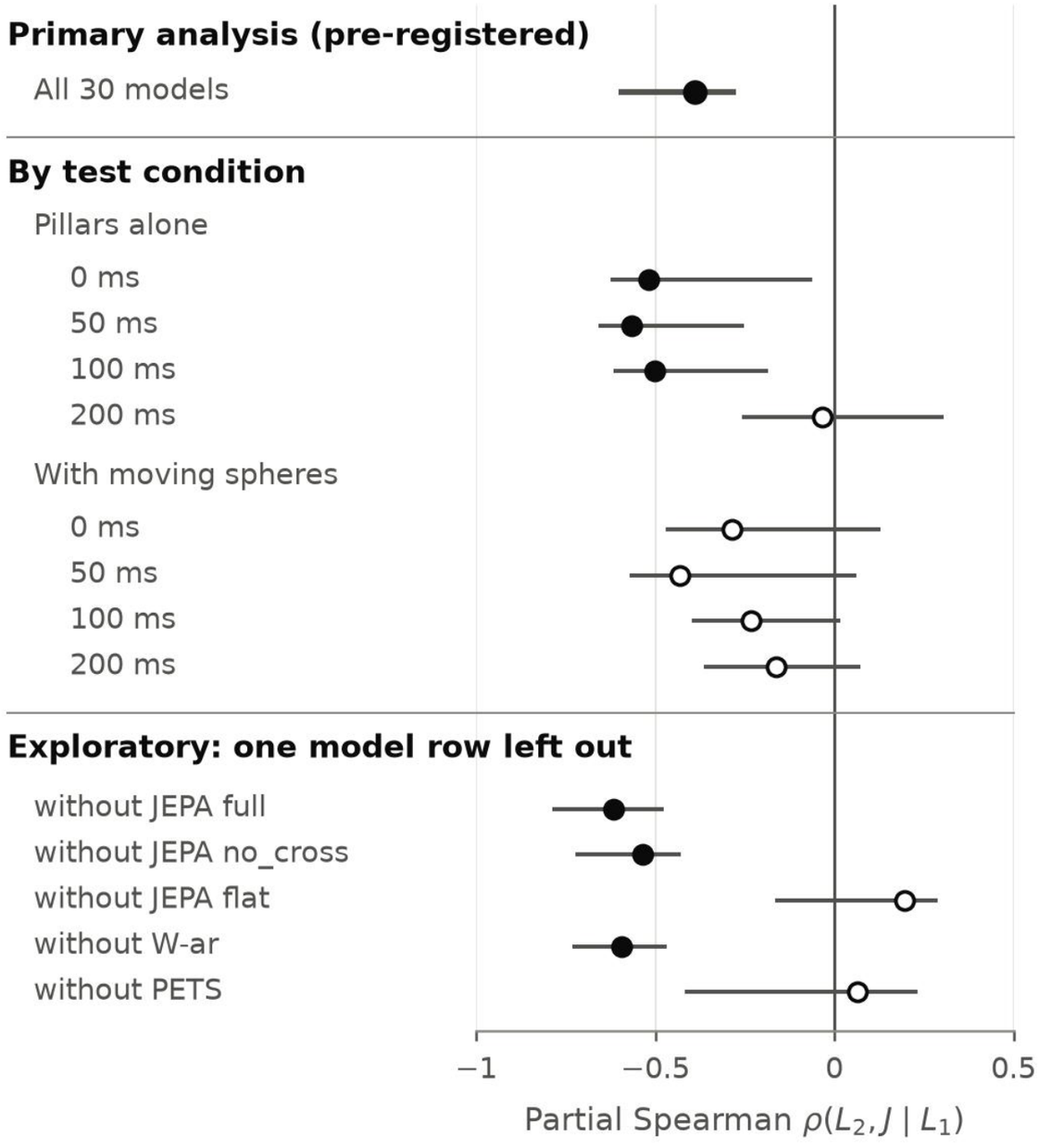
分组标题单独成行，加粗并配分隔线；具体方法名缩进。每根条形都对应一行方法名及训练数据量，取消跨两行居中。Training episodes 列标题与数据量列对齐。

横轴为全部测试任务的比例，包含成功任务。保留全部 17 行和 5 类终止原因，不删减 500/2,000 数据量，不新增不确定性区间或逐段标签。

Original



Revised (local review)



Points: estimates; lines: 95% bootstrap intervals.

Filled: interval below 0; open: interval includes 0.

Table 1：原版与本地改进版

原版

TABLE I
THE ARENA AND ITS EVALUATION

Arena	
Team	three Crazyflie 2 on 0.6 m tension-only cables, one load
Plant	MuJoCo and onboard Lee controllers, both at 500 Hz
Load mass	10–40 g, drawn per layout and hidden; nominal 25 g
Mission	goal 3.2 m ahead within 20 s; success once the load stays within 0.15 m of it, below 0.2 m/s, for 0.5 s
Obstacles	four pillars; in half the conditions also two spheres at 0.25–0.45 m/s
Decisions	10 Hz, on the state from 0, 50, 100 or 200 ms ago
Planner	MPPI: 512 candidate plans of 15 decisions (1.5 s), setpoint moves of at most 0.12 m per axis
Mass estimation	16 physics models on a 2 g grid; the best over the last 0.5 s plans
Test set	50 layouts $\times$ 8 conditions $=$ 400 episodes
Learned models	5 rows $\times$ 2 data sizes $\times$ 3 seeds $=$ 30, with 1.28–1.37 M parameters
Training data	2,000 episodes (170,304 decisions, 4.7 h); 100 for validation
Policies	CrazyMARL 2×10^9 steps per seed, 8,192 parallel environments (official 16,384); TD-MPC2 and DreamerV3 2×10^6 steps per seed
Evaluation	
Questions	every fifth decision of U's test flights, 8,660 in all
Layer 2 changes	eight per aircraft, Gaussian, 0.04 m per axis and decision; median true effect 6.2 cm on the load and 2.7 cm on another aircraft
Cost unit	about 1.7 cm cut into a safety margin, or 0.6 m less progress
Layer 3 questions	the 89% at which U's plan is within one cost unit of its best candidate
Intervals	95%, from 10,000 bootstrap resamples of the 50 test layouts
Power	67–79% for a partial correlation of 0.43–0.48 (simulated)

改进版

TABLE I
TASK, PLANNING, AND EVALUATION SETTINGS

Task and environment	
Team	Three Crazyflie 2 quadrotors; one load; 0.6 m tension-only cables
Load mass	10–40 g per layout, hidden; nominal mass 25 g
Obstacles	Four pillars, with or without two moving spheres (0.25–0.45 m/s)
Mission	Goal 3.2 m ahead; time limit 20 s
Success criterion	Load within 0.15 m of the goal, below 0.2 m/s, continuously for 0.5 s
Control and planning	
Simulation/control	MuJoCo and onboard Lee controllers, both at 500 Hz
Decision rate	10 Hz
Observation delay	0, 50, 100, or 200 ms
Shared planner	MPPI; 512 candidate plans per decision
Planning horizon	15 decisions (1.5 s)
Setpoint moves	At most 0.12 m per axis
Data and evaluation	
Training budgets	500 / 2,000 episodes; 500 uses the first quarter of the same dataset
Full training set	170,304 decisions (4.7 h); 100 separate validation episodes
Main model set	5 model rows $\times$ 2 budgets $\times$ 3 new seeds $=$ 30 models; 1.28–1.37 M parameters
Test coverage	50 layouts $\times$ 2 obstacle settings $\times$ 4 delays $=$ 400 episodes per model
Probe states	Every fifth decision of U's test flights; 8,660 states in total
Uncertainty	95% intervals; 10,000 bootstrap resamples of the 50 test layouts

Table 2：原版与本地改进版

原版

TABLE II
RESULTS ON THE 400 TEST EPISODES

Row	J (%) [95% CI]	L_1 (cm)	L_2 (cm)	L_3 (%)
Reference rows, shared MPPI				
U (oracle)	76.3 [65.0, 86.8]	0	0	0
Physics + mass est.	77.8 [67.5, 87.0]	0.57	0.47	3.7
Physics (25 g)	53.3 [42.5, 64.3]	8.42	3.62	33.8
Learned world models, shared MPPI				
Example MLP	15.3 [9.3, 22.3]	12.81	4.67	51.9
JEPA `full`, 500 ep.	34.9 [26.5, 43.6]	10.84	4.07	47.8
JEPA `full`, 2,000 ep.	50.4 [40.3, 60.3]	9.01	3.56	47.6
JEPA `no_cross`, 500 ep.	30.0 [22.6, 37.8]	10.67	3.82	46.7
JEPA `no_cross`, 2,000 ep.	49.4 [39.8, 59.0]	8.80	3.48	46.1
JEPA `flat`, 500 ep.	12.2 [7.1, 18.3]	12.59	4.41	51.4
JEPA `flat`, 2,000 ep.	16.2 [11.4, 21.5]	11.20	4.25	56.4
W-ar, 500 ep.	45.3 [36.2, 54.3]	9.46	3.44	43.7
W-ar, 2,000 ep.	46.8 [37.8, 55.5]	8.55	3.07	41.4
PETS, 500 ep.	17.2 [11.4, 23.7]	13.24	3.92	47.8
PETS, 2,000 ep.	23.3 [16.4, 30.9]	12.53	3.72	46.6
Policies without the shared planner				
CrazyMARL	40.6 [30.0, 51.5]	–	–	–
TD-MPC2	0.0[a]	–	–	–
DreamerV3	0.0[a]	–	–	–

J: success rate, higher is better; L_1–L_3: layers 1–3 (Section IV-B), lower is better. Brackets: 95% bootstrap interval over layouts. Rows with three seeds pool J over the seeds' 1,200 episodes and average L_1–L_3 over the three models; ep. = training episodes. [a]No success with any of the three seeds. Fig. 5 shows every seed of the world models.

改进版

TABLE II
BENCHMARK RESULTS ACROSS EIGHT TEST CONDITIONS

Method	Train ep.	J (%) ↑ [95% CI]	L_1 (cm) ↓	L_2 (cm) ↓	L_3 (%) ↓
Reference rows, shared MPPI					
U (oracle)	–	76.3 [65.0, 86.8]	0	0	0
Physics + mass est.	–	77.8 [67.5, 87.0]	0.57	0.47	3.7
Physics (25 g)	–	53.3 [42.5, 64.3]	8.42	3.62	33.8
Learned world models, shared MPPI					
Example MLP	–	15.3 [9.3, 22.3]	12.81	4.67	51.9
JEPA `full`	500	34.9 [26.5, 43.6]	10.84	4.07	47.8
JEPA `full`	2,000	50.4 [40.3, 60.3]	9.01	3.56	47.6
JEPA `no_cross`	500	30.0 [22.6, 37.8]	10.67	3.82	46.7
JEPA `no_cross`	2,000	49.4 [39.8, 59.0]	8.80	3.48	46.1
JEPA `flat`	500	12.2 [7.1, 18.3]	12.59	4.41	51.4
JEPA `flat`	2,000	16.2 [11.4, 21.5]	11.20	4.25	56.4
W-ar	500	45.3 [36.2, 54.3]	9.46	3.44	43.7
W-ar	2,000	46.8 [37.8, 55.5]	8.55	3.07	41.4
PETS	500	17.2 [11.4, 23.7]	13.24	3.92	47.8
PETS	2,000	23.3 [16.4, 30.9]	12.53	3.72	46.6
Policies without the shared planner					
CrazyMARL	–	40.6 [30.0, 51.5]	–	–	–
TD-MPC2	–	0.0[a]	–	–	–
DreamerV3	–	0.0[a]	–	–	–

400 test episodes per model. For three-seed methods, J pools 1,200 episodes; L_1–L_3 are seed means. Brackets: 95% bootstrap CIs over 50 layouts. Train ep.: offline training episodes; –: not reported or not applicable. [a]Zero successes across all three seeds.